

A Chinese herbal medicine Ermiao wan reduces serum uric acid level and inhibits liver xanthine dehydrogenase and xanthine oxidase in mice

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Abstract

Ermiao wan, which is composed of phellodendri cortex and atractylodis rhizome, is described as eliminating heat, excreting dampness and anti-edema prescription in traditional Chinese medical literatures including Danxi's Experiences in Medicine and State Pharmacopoeia of People's Republic of China. So it is being used clinically in the treatment of gout and hyperuricemia in China. In the present study, the water extracts of Ermiao wan and phellodendri cortex at 840 and 480 mg/kg/day orally for 7 days were demonstrated to possess *in vivo* potent hypouricemic effects both in hyperuricemic mice pretreated with oxonate and in normal mice, respectively. In the hyperuricemic animals, the effect of Ermiao wan was equal to that of the reference drug allopurinol (at 10 mg/kg/day orally for 7 days), but in the normal mice, the former was weaker than latter. In addition, both Ermiao wan and phellodendri cortex were found to have *in vivo* relatively inhibitory effects on mouse liver xanthine dehydrogenase (XDH) and xanthine oxidase (XO) activities at the same dose described above. These inhibitory effects were weaker than that observed for allopurinol. Atractylodis rhizome at 340 mg/kg/day orally for 7 days did not show any effects on the above experiments. These results suggested that atractylodis rhizomes assisted and enhanced the effect of phellodendri cortex on reduction of serum uric acid level in hyperuricemic mice, and hypouricemic effects of Ermiao wan and phellodendri cortex may be achieved by other mechanism partly instead of the XDH and XO inhibition.

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1. Introduction

Gout and hyperuricemia are metabolic disorders associated with abnormal amounts of uric acid in the body, resulting in the deposition of urate crystals in the joints and kidneys, causing inflammation as well as gouty arthritis and uric acid nephrolithiasis (Kramer and Curhan, 2002; Tomita et al., 2000). Xanthine dehydrogenase (XDH) and xanthine oxidase (XO), which are present in significant concentration only in liver and intestine, oxidizes hypoxanthine and xanthine to uric acid in the purine catabolic pathway. Treatment and prevention of gout and hyperuricemia entail to reduce serum uric acid concentration in the body. XDH and XO inhibitors are available to block the final step in uric acid synthesis, reducing the production of uric acid, as well

as anti-inflammatory agents to relieve the symptoms of the disease (Kelley, 1991; Star and Hochberg, 1993; Wortmann, 1993). Allopurinol is the only inhibitor of the enzyme in clinical use, but its use can result in a number of adverse side effects, such as allergic and hypersensitivity reactions, nephropathy, and enhancement of 6-mercaptopurine toxicity (Hande et al., 1984; Urban et al., 1995; Sumi and Wada, 1996). Thus, the development of new hypouricemic agents of greater effectiveness and safety is highly warranted.

A Chinese herbal medicine Ermiao wan, which is composed of the phellodendri cortex and atractylodis rhizome, has long been used in traditional medicine. According to Dan-Xi-Xin-Fa (Zhu Zhenhen, Yuan dynasty: a Chinese medicine book named Danxi's Experiences in Medicine), Ermiao wan was described to treat acute gout through eliminating heat and excreting dampness in terms of traditional Chinese medicine. Ermiao wan is recorded in State Pharmacopoeia of People's Republic of China at all times for the successful treatment of gout and hyperuricemia. Despite the

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facts that phellodendri cortex and atractylodis rhizome have shown antioxidant and anti-inflammatory activities (Endo et al., 1979; Yamahara et al., 1990; Resch et al., 1998; Lee et al., 2000), it is not clear whether Ermiao wan can reduce serum uric acid level in a hyperuricemia model and inhibit XDH and XO activities. The aim of the present study was to evaluate the effects of Ermiao wan in vivo on reduction of serum uric acid level in hyperuricemic mice and the activities of mouse liver XDH and XO.

2. Materials and methods

2.1. Reagents

Xanthine, nicotinamide adenine dinucleotide (NAD^+), uric acid and allopurinol were purchased from Sigma (St. Louis, MO, USA). Potassium oxonate was purchased from Aldrich Inc. Other reagents were of analytical grades.

2.2. Plant material

Ermiao wan was composed of the following air-dried powdered raw materials: phellodendri cortex, 15 g (*Phellodendron chinense* Schneid. (Rutaceae)); rhizome atractylodis, 15 g (*Atractylodes lancea* (Thunb.) DC. (Asteraceae)). The two plants were purchased from Medicinal Materials Co. of Jiangsu Province and authenticated by Associate Prof. L.X. Zhang, School of Life Sciences, Nanjing University, respectively. The voucher specimens (NU-18010 and NU-21761) have been deposited in the Herbarium of Nanjing University, Nanjing University, Nanjing 210093, P.R. China.

2.3. Preparation of the aqueous extract

Powdered barks of *Phellodendron chinense* (150 g) and rhizomes of *Atractylodes lancea* (150 g) were extracted with boiling water (1500 ml) for 45 min and filtered through a two-layer mesh and concentrated on a boiling bath to obtained residues (yields of 18.5 and 13.4%, respectively). In this experiment, Ermiao wan extract was prepared by extracting the above crude drug mixture (300 g) with boiling water (3000 ml) for 45 min, filtrating the solution and evaporating the solvent to get residue (yield of 16.2%). These extracts were stored at 4 °C before use.

The reference drug allopurinol was suspended in distilled water at the concentration of 1 mg/ml.

2.4. Animals

Male ICR mice (26–30 g) were purchased from the Laboratory Animal Center (Nanjing, Jiangsu Province, China). They were allowed at least 1 week to adapt to their environment before used for experiments. They were housed at room temperature with a regular light/dark cycle, and given

a standard chow and water ad libitum for the duration of the study.

2.5. Animal model of hyperuricemia in mice

In vivo drug-induced hyperuricemic animal model was adopted (Stavric et al., 1975; Hall et al., 1990). The uricase inhibitor potassium oxonate was used to induce hyperuricemia in mice. Briefly, mice were intraperitoneally injected with potassium oxonate at 250 mg/kg 1 h before the final tested drug administration to increase the serum urate level. Whole blood samples were collected from mice by tail vein bleeding 1 h after final administration. The blood was allowed to clot for approximately 1 h at room temperature and then centrifuged at $3500 \times g$ for 5 min to obtain the serum. The serum was stored at -20°C until assayed. Serum uric acid was determined with the phosphotungstic acid method (Carroll et al., 1971).

2.6. Enzyme preparation

Mice were decapitated and the liver rapidly excised and homogenized in 5 vol. of 50 mM ice-cold phosphate buffer (pH 7.5) using a motor-driven Teflon glass homogenizer. The homogenate was then centrifuged for 10 min at $1500 \times g$ at 4 °C and resulting supernatant fraction was further centrifuged at $100,000 \times g$ for 60 min at 4 °C in a preparative centrifuge. The supernatant obtained after the latter centrifugation was used for the assays of XDH and XO activities.

2.7. Assays of mouse liver XDH and XO activities

The activities of XDH and XO were assayed by monitoring uric acid formation using a spectrophotometric method as described elsewhere (Hall et al., 1990; Sugawara et al., 1999). The reaction mixture contained xanthine (as the substrate, final concentration 50 μM), EDTA (final concentration 0.08 mM). The reaction was started by adding 100 μl of the supernatant in a phosphate buffer solution (pH 7.5, 200 mM). The mixture (total 5.5 ml) was incubated for 30 min at 37 °C. Prior to the measurement of uric acid production by measuring the UV absorbance at 290 nm, the reaction was terminated by the addition of 0.58 M HCl (0.5 ml). Reaction buffer contained 200 μM NAD^+ as the electron acceptor for the XDH assay. XDH and XO activities were expressed as nmol/min per mg protein. Each assay was performed in triplicate. Protein concentration was determined by the Lowry method (Lowry et al., 1951) using bovine serum albumin as the standard.

2.8. Drug administration

Food, but not water, was withdrawn from the animals 1.5 h prior to drug administration. Water extracts of Ermiao wan, barks of *Phellodendron chinense*, and rhizomes of *Atractylodes lancea* and allopurinol at various concentrations were

Table 1

Effects of the water extracts of Ermiao wan, phellodendri cortex and atractylodis rhizome on serum urate levels in mice treated with the uricase inhibitor, potassium oxonate^a

Treatment	Dose (mg/kg)	n	Uric acid (mg/dl) mean $\pm$ S.E.M.	Inhibition ^b (%)
Vehicle	–	12	3.24 $\pm$ 0.41	–
PO alone	–	12	5.73 $\pm$ 0.15 ^{##}	–
PO + EMW	840	12	3.39 $\pm$ 0.14 ^{***}	40.8
PO + PC	480	12	4.14 $\pm$ 0.10 ^{**}	27.7
PO + AR	340	12	5.06 $\pm$ 0.15 [#]	11.7
PO + AP	10	12	3.32 $\pm$ 0.11 ^{***}	42.1

^a PO: potassium oxonate; EMW: Ermiao wan; PC: phellodendri cortex; AR: atractylodis rhizome; AP: allopurinol.

^b Inhibition (%) = 100((PO alone – experiment)/PO alone).

[#] $P < 0.01$ when compared with vehicle control groups.

^{##} $P < 0.001$ when compared with vehicle control groups.

^{**} $P < 0.01$ when compared with PO groups.

^{***} $P < 0.001$ when compared with PO groups.

Table 2

Effects of the water extracts of Ermiao wan, phellodendri cortex and atractylodis rhizome on serum urate levels in mice untreated with the uricase inhibitor, potassium oxonate^a

Treatment	Dose (mg/kg)	n	Uric acid (mg/dl) mean $\pm$ S.E.M.	Inhibition (%)
Vehicle	–	12	3.68 $\pm$ 0.08	–
EMW	840	12	2.91 $\pm$ 0.10 [*]	20.9
PC	480	12	2.78 $\pm$ 0.16 ^{**}	24.5
AR	340	12	3.21 $\pm$ 0.14	12.8
AP	10	12	2.35 $\pm$ 0.11 ^{***}	27.2

^a EMW, Ermiao wan; PC, phellodendri cortex; AR, atractylodis rhizome; AP, allopurinol.

^{*} $P < 0.05$ when compared with vehicle control group.

^{**} $P < 0.01$ when compared with vehicle control group.

^{***} $P < 0.001$ when compared with vehicle control group.

dissolved in propyleneglycol/water (50/50). The volume of the suspension administered was based on body weight measured immediately prior to each dose, respectively. All drugs were given orally or intraperitoneally once daily at 14:00.

As showed in Table 1, vehicle control and potassium oxonate hyperuricemia groups were orally administered with 0.9% saline solution for 7 days, respectively. The other four groups received extracts of Ermiao wan (840 mg/kg), barks

of *Phellodendron chinense* (480 mg/kg), and rhizomes of *Atractylodes lancea* (340 mg/kg) and allopurinol (10 mg/kg), respectively for 7 days, respectively. The serum uric acid assay started in potassium oxonate pretreated and untreated mice 1 h after final administration, respectively.

As showed in Tables 2 and 3, the vehicle control group was administered with 0.9% saline solution for 7 days. The other four groups received extracts of Ermiao wan (840 mg/kg), barks of *Phellodendron chinense* (480 mg/kg), and rhizomes of *Atractylodes lancea* (340 mg/kg) and allopurinol (10 mg/kg), respectively for 7 days. The serum uric acid and the liver enzyme activity assays started 1 h after final administration, respectively.

2.9. Statistical analysis

All data were expressed as the mean $\pm$ standard error of the mean (S.E.M.) and statistical analysis was performed with an analysis of variance followed by Student's *t*-test. The results were considered significant if the probability of error was $<5\%$.

3. Results

3.1. Hypouricemic activities in hyperuricemic mice

The hypouricemic effects of water extracts of the Ermiao wan, phellodendri cortex and atractylodis rhizome on potassium oxonate-induced hyperuricemia in mice were showed in Table 1. In normal mice, serum uric acid level was 3.24 ± 0.41 mg/dl. In hyperuricemic animals, serum uric acid level was elevated to 5.73 ± 0.15 mg/dl (176.9% of control) 2 h after intraperitoneal injection of potassium oxonate. In contrast, serum uric acid levels of the hyperuricemic group orally treated with water extracts of Ermiao wan and phellodendri cortex for 7 days were significantly lowered to 40.8 and 27.7% of the hyperuricemic group, respectively. Thus, both Ermiao wan and phellodendri cortex were able to induce an immediate decrease in serum uric acid levels in hyperuricemic mice. Moreover, the serum uric acid level of the hyperuricemic mice treated with the water extract of Ermiao wan was not significantly different from that of vehicle

Table 3

Effects of the water extracts of Ermiao wan, phellodendri cortex and atractylodis rhizome on xanthine dehydrogenase and xanthine oxidase activities in the mouse liver^a

Treatment	Dose (mg/kg)	n	XDH (nmole uric acid/min per mg protein) mean $\pm$ S.E.M.	XO (nmole uric acid/min per mg protein) mean $\pm$ S.E.M.	XDH inhibition (%)	XO inhibition (%)
Vehicle	–	12	2.51 $\pm$ 0.13	2.31 $\pm$ 0.08	–	–
EMW	840	12	1.78 $\pm$ 0.13 [*]	1.83 $\pm$ 0.11 [*]	29.1	20.8
PC	480	12	1.65 $\pm$ 0.06 [*]	1.67 $\pm$ 0.03 [*]	34.3	27.7
AR	340	12	2.17 $\pm$ 0.18	2.37 $\pm$ 0.12	13.5	–
AP	10	12	1.27 $\pm$ 0.11 [*]	1.32 $\pm$ 0.04 [*]	49.4	42.9

^a EMW: Ermiao wan; PC: phellodendri cortex; AR: atractylodis rhizome; AP: allopurinol.

^{*} $P < 0.001$ when compared with vehicle control group.

mice. However, serum uric acid level of the hyperuricemic group orally treated with water extract of *atractylodis rhizome* for 7 days was not affected. The hypouricemic effect of *Ermiao wan* was more than that of *phellodendri cortex*. In the study, the reference drug *allopurinol* also significantly lowered serum uric acid level to normal value in mice.

3.2. Hypouricemic activities in normal mice

Normal mice were orally treated with the same extracts for 7 days to examine the possibility that the hypouricemic effects of these extracts observed in the hyperuricemic mice could also yield a beneficial effect in normal mice. Initial mean of serum uric acid level in normal mice was 3.68 ± 0.08 mg/dl (Table 2). The water extracts of *Ermiao wan* and *phellodendri cortex* effectively produced decrease in serum uric acid level in the normal animals (2.91 ± 0.10 and 2.78 ± 0.16 mg/dl, respectively), exhibiting 20.9 and 24.5% inhibition of the control group, respectively. However, the water extracts of *atractylodis rhizome* did not affect in serum uric acid level in the normal animals (3.21 ± 0.14 mg/dl). *Allopurinol* also significantly lowered serum uric acid level in normal animals, exhibiting to be more potent than *Ermiao wan* and *phellodendri cortex*.

3.3. Liver XDH and XO inhibitory activities in mice

In order to perform detailed investigations of the hypouricemic properties of these water extracts, we need to assay *in vivo* mouse liver XDH and XO activities. The effects of water extracts of *Ermiao wan*, *phellodendri cortex* and *atractylodis rhizome* *in vivo* on mouse liver XDH and XO activities were showed in Table 3. The XDH and XO activities in vehicle group were 2.51 ± 0.13 and 2.31 ± 0.08 nmol/min per mg protein, respectively. The XDH activities in *Ermiao wan* and *phellodendri cortex* groups were 1.78 ± 0.13 and 1.65 ± 0.06 nmol/min per mg protein, respectively. The XO activities in *Ermiao wan* and *phellodendri cortex* groups were 1.83 ± 0.11 and 1.67 ± 0.03 nmol/min per mg protein, respectively. Thus, *Ermiao wan* and *phellodendri cortex* caused inhibition by 29.1 and 34.3% towards XDH, 20.8 and 27.7% towards XO as compared to the vehicle group, respectively. In contrast, *atractylodis rhizome* had no effect on the activities of XDH and XO (2.17 ± 0.18 nmol/min per mg protein and 2.37 ± 0.12 nmol/min per mg protein, respectively). *Allopurinol* inhibited both XDH (49.4% inhibition) and XO (42.9% inhibition) activities at the dose of 10 mg/kg, showing to be more potent than that of *Ermiao wan* and *phellodendri cortex*.

4. Discussion

Gout is a common metabolic disorder in human, reportedly afflicting more than 2 million men and women in the United States alone (Kramer and Curhan, 2002). The disease

is also rapidly rising in China (Li et al., 1997) due probably to recent changes in dietary habits. Therapeutic approaches for gout treatment include the use of anti-inflammatory agents for symptomatic relief and of XDH and XO inhibitors, which block the synthesis of uric acid from purines.

In the present study, it was demonstrated that 7-day oral administration of *Ermiao wan* was able to significantly reduce the serum urate levels in hyperuricemic mice. There was really no significant difference between *Ermiao wan* and *allopurinol* in this hypouricemic effects. Direct comparison of *Ermiao wan* and its composed plants, *phellodendri cortex* and *atractylodis rhizome* in the hyperuricemic model indicated that *Ermiao wan* was more effective than that of *phellodendri cortex* (serum uric acid level reduction of 40.8 and 27.7% as compared with oxonate-pretreated group, for the former and latter, respectively), whereas, *atractylodis rhizome* did not produce hypouricemic effect. Based on the theory of traditional Chinese medicine, *phellodendri cortex* is a chief ingredient, while *atractylodis rhizome* is an adjuvant ingredient in *Ermiao wan* prescription, respectively. The combined use of the two drugs (*Ermiao wan*) could produce mutual enhancement. In fact, our data indicated the hypouricemic effect of *atractylodis rhizome* was enhanced when *phellodendri cortex* was present in hyperuricemic animal model.

Phellodendri cortex and *atractylodis rhizome* have been described previously to possess weaker XO inhibitory activity *in vitro* on a commercial enzyme preparation (Kong et al., 2000). However, in the present study, the extracts of *Ermiao wan* and *phellodendri cortex* were found to have *in vivo* relatively inhibitory activities of mouse liver XDH and XO. It was noteworthy to mention that some compounds such as pteridines, which was not xanthine inhibitors *in vitro*, have also been reported to possess *in vivo* hypouricemic effects (Ferrand et al., 1992). On the other hand, *Ermiao wan* was less efficient than *phellodendri cortex*, and *atractylodis rhizome* had no detectable change in XDH and XO activities. *Allopurinol* was more potent than *Ermiao wan* and *phellodendri cortex*. It should be noted that the magnitude of the hypouricemic action did parallel the reduction in the enzyme activities. Based on this observation, it is postulated that hypouricemic effects of *Ermiao wan* and *phellodendri cortex* in hyperuricemic mice were achieved by other mechanism partly instead of the XDH and XO inhibition *in vivo*.

Gout and hyperuricemia are very complicated. Our findings regarding the *in vivo* hypouricemic actions of *Ermiao wan* indicate that traditional Chinese medicine, besides anti-inflammatory agents and XDH and XO inhibitors, can form a viable natural drug of such a cocktail therapeutic approach. This is not surprising in view of the many side-effects reported of *allopurinol*. *Ermiao wan* has long been used in China to successfully treat gout and hyperuricemia. Although its mechanism action of the hypouricemic effects is not fully understood, it does appear to be a viable substitute for *allopurinol*.

Several groups of bioactive natural products have been identified from the two herbs in Ermiao wan, the main groups being: alkaloids (e.g. berberine, jatrorrhizine, magnoflorine, palmatine and phellodendrine isolated from phellodendri cortex (Jiangsu College of New Medicine, 1977), terpenes and polyacetylenic compounds (e.g. hinesol, beta-eudesmol, elemol, atractylodin, atractylodinol, acetylactylodinol, 4,6,12-tetradecatriene-8,10-diyne-1,3,14-triol, 2-[(2'E)-3',7'-dimethyl-2', 6'-octadienyl]-4-methoxy-6-methylphenol (3Z,5E,11E)-tridecatriene-7,9-diynyl-1-O-(E)-ferulate and (4E,6E,12E)-tetradecatriene-8,10-diyne-1,3-diyl diacetate) (Guo et al., 2002; Nakai et al., 2003), sesquiterpenoid glycosides (acetylene, eudesmane and guaiane-type glycosides) (Kitajima et al., 2003), fatty acids (e.g. linoleic acid, oleic acid, and palmitic acid) and polysaccharides (Inagaki et al., 2001) isolated from atractylodis rhizome. Berberine is major component of phellodendri cortex, and has many biological activities, such as bactericidal activity, anti-cholera toxin effect, anti-ulcer, anti-inflammatory properties, stimulative effect of bile secretion or bilirubin discharge and inhibitory effect on xanthine oxidase (Uchiyama et al., 1989; Chang et al., 1994; Miura et al., 1997). Magnoflorine exhibited lipoxygenase inhibition and antioxidant properties (Misik et al., 1995; Hisatomi et al., 2000). Phellodendrine was expected to be a valuable new type of immunosuppressor against the cellular immune response (Mori et al., 1995) and have antinephritic effect (Hattori et al., 1992). Sesquiterpenoid components atractylon, beta-eudesmol, and hinesol exerted significant liver-protective effects (Kiso et al., 1983). Beta-eudesmol improved the tetanic contraction (Chiou et al., 1995) and potentiated the neuromuscular blocking effect of succinylcholine (Kimura et al., 1994). Hinesol strongly inhibited H^+ , K^+ ATPase activity and have an anti-gastric ulcer activity (Satoh et al., 2000). Elemol also was confirmed to have antiulcer activity (Matsunaga et al., 2000). Recently, it was found that the four polyacetylenic compounds atractylodin, atractylodinol, acetylactylodinol and 4,6,12-tetradecatriene-8, 10-diyne-1,3,14-triol, but not the two sesquiterpens hinesol and beta-eudesmol, improved the delayed gastric emptying. However, the polyacetylenic compounds contributed to this letter activity in a similar potency (Nakai et al., 2003). The polyacetylenes atractylodochromene, 2-[(2E)-3,7-dimethyl-2,6-octadienyl]-6-methyl-2,5-cyclohexadiene-1,4-dione, 2-[(2'E)-3',7'-dimethyl-2',6'-octadienyl]-4-methoxy-6-methylphenol exhibited potent inhibitory activities on 5-lipoxygenase (5-LOX) and cyclooxygenase-1 (COX-1) (Resch et al., 1998, 2001). Atractylon and the coumarin osthol turned out to be moderate but selective 5-lipoxygenase inhibitors (Resch et al., 1998). Moreover, linoleic acid, oleic acid and palmitic acid showed 5-LOX-/COX-1 inhibitory actions in vitro (Resch et al., 2001). The polysaccharide component of F-2-2 played a major role in the protective activity against Candida-infected mice (Inagaki et al., 2001; Yu et al., 1998). These constituents may be present in different amounts in

the extracts of Ermiao wan and it is far from established which constituent(s) are responsible for the therapeutic effects of the extracts. Further investigations are warranted to identify the active principle(s) of the extracts from Ermiao wan and the two herbs, responsible for the observed hypouricemic effects.

In conclusion, the objective of the present study was to test the hypouricemic effects of the Ermiao wan, which is used in traditional Chinese medicinal to treat gout and hyperuricemia. Water extracts of Ermiao wan and phellodendri cortex possessed in vivo potent hypouricemic effects both in hyperuricemic mice pretreated with oxonate and in normal mice, respectively. In addition, both Ermiao wan and phellodendri cortex were found to have in vivo relatively inhibitory effects on mouse liver XDH and XO activities. These results suggested that atractylodis rhizomes assisted and enhanced the effect of phellodendri cortex on reduction of serum uric acid level in hyperuricemic mice, and hypouricemic effects of Ermiao wan and phellodendri cortex may be achieved by other mechanism partly instead of the XDH and XO inhibition.

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